ENVS 423L/EPS 522 Problem Set #1

Katie Markovich

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Calculations, methods, and assumptions should be spelled out so that your work can be reproduced. All work should be shown to receive full credit.

Upload a word document or PDF of your solutions/answers by midnight on Thursday, August 28 to Canvas.

1 Introduction

The purpose of this problem set is to reinforce your understanding of some basic hydrologic concepts.

1.1 Unit Conversion

Complete the following unit conversions (10 pts)

Convert	to	to	to
Weight density of water 1.00 g_f/cm	______ kg_f/m^3	______ lb/ft^3	______ N/m^3
Atmospheric pressure 14.7 lb/ft^2	______ $dyne/cm^2$	______ N/m^2	______ kPa
Dynamic viscosity of water 1.80×10^{-2} $dyne \cdot s/cm^2$	______ $N \cdot s/m^2$	______ $lb \cdot s/ft^2$	______ $Pa \cdot s$
Velocity 14 cm/s	______ ft/s	______ m/d	______ mi/hr
Discharge 653 ft^3/s	______ m^3/s	______ gal/min	______ gal/d
Rainfall-runoff conversion 1.00 in rain on 1.00 mi^2	______ ft^3	______ gal	______ m^3
Rainfall-runoff conversion 1.00 in rain on 1.00 mi^2 for 1.00 hr	______ ft^3/s	______ L/s	______ m^3/s
Water-balance conversion 1.00 in rain on 1.00 mi^2 in 1.00 yr	______ ft^3/s	______ L/s	______ m^3/s
Area 1.00 mi^2	______ m^2	______ km^2	______ ha
Temperature 12°F	______ °C	______ K	
Temperature 63.2°F	______ °C	______ K	
Temperature 14.7°C	______ °F	______ K	
Temperature 20°C	______ °F	______ K	
Temperature difference of 9.6 F°	______ C°	______ K	
Temperature difference of 12.9 C°	______ F°	______ K	
Heat capacity of water 1.00 cal/g · °C	______ J/kg · K	______ $BTU/lb_m \cdot °F$	

1.2 Dimensional Analysis, part 1

What are the dimensions of dynamic viscosity (μ) and kinematic viscosity (ν)? (10 pts)

The Reynolds number (Re) is a quantity that describes whether fluid flow is laminar or turbulent. Given Equation 2, where ρ is density, u is velocity, μ is dynamic viscosity, and L is length, what are the dimensions of Re? (10 pts)

$$Re = \frac{\rho u L}{\mu} \tag{1}$$

1.3 Capillary Rise

Calculate the expected capillary rise for a glass tube with radius of 0.5 mm and for water at standard temperature and pressure (STP). Assume $cos(\theta_c)$ is 1 (for a contact angle of $\tilde{0}$ degrees, which is common for water and most soil materials). (10 pts)

For the same radius and contact angle, what is the expected rise for water at 30 degrees C? Recall both surface tension and weight density are temperature-dependent. (10 pts)

1.4 Water Balance

A lake with drainage area A_d (L^2) has a surface area of A_L (L^2) and an average depth of z_L (L). The long-term average rates of precipitation and evapotranspiration for the land portion of the drainage basin are p_D and e_D, respectively. For the lake, average rates of precipitation and evaporation are p_L and e_L, respectively. The average rate of stream outflow from the lake is Q. Assume there is no groundwater inflow or outflow to the basin or lake.

Using these symbols and assuming a consistent set of units:

- Write the water-balance equation for the drainage basin. (10 pts)

- Write the water-balance equation for the lake. (10 pts)

1.5 Water Balance, cont'd.

Given the following data for a drainage basin and lake:

Drainage basin: A_D=940 km^2, p_D=1,240 mm/yr, e_D= 500 mm/yr

Lake: A_L=180 km^2, p_D=952 mm/yr, Q= 15.1 m^3/s, z_L=18 m

- Use the drainage basin and lake water balance equations above to estimate lake evaporation. Do they differ? (10 pts)

- Compare the estimates of e_L above to the given value of e_D. What are possible explanations for the difference? (10 pts)

- Compare the magnitude of the inflows to the lake from the drainage-basin runoff to lake precipitation. What are the water quality implications of these numbers? (10 pts)

1.6 Extra Credit: Where does your water come from?

Pick your hometown (or any other place you have lived) and describe where drinking water for that town comes from. If there are multiple sources of water, describe them all. What are the current/future risks to this drinking water source (10 pts)